

VisionRoot - Week 1 – AI & LLM FOUNDATION

Practical task

1. What Does an AI Engineer Do in a Software Company?

An AI Engineer is a software professional who develops, integrates, deploys, and maintains artificial intelligence capabilities within real-world software applications. Unlike working only with an AI model, an AI Engineer focuses on turning AI capabilities into reliable, usable, and maintainable products.

In a software company, an AI Engineer may work with machine learning models, Generative AI systems, Large Language Models (LLMs), APIs, databases, and backend services. They are responsible for understanding the business problem and selecting an appropriate AI approach to solve it. For example, an AI Engineer may develop a customer-support assistant, document-processing system, recommendation system, fraud-detection solution, or an AI-powered internal productivity tool.

A typical AI application contains several components. A user interacts with a frontend application, which sends a request to a backend service. The backend communicates with an AI service, which prepares the required data and prompt and interacts with an AI model. The model generates a response, which is processed by the application and returned to the user. Therefore, an AI model is only one component of a complete AI-powered software product.

AI Engineers are also responsible for engineering controls around AI systems. These include data validation, prompt design, API management, authentication, security, error handling, logging, evaluation, and monitoring. These controls are important because AI models can produce incorrect, incomplete, or unexpected outputs. A production AI system therefore needs to be tested and monitored rather than relying only on the model's capabilities.

Python is an important programming language in AI Engineering because it provides libraries and frameworks for data processing, machine learning, APIs, automation, and AI model integration. AI Engineers may also work with technologies such as FastAPI, REST APIs, databases, cloud platforms, Git, and model-provider SDKs.

Another important responsibility is evaluating AI applications. Engineers need to determine whether an AI system provides useful and reliable results. Evaluation may include checking accuracy, relevance, hallucinations, latency, cost, safety, and user feedback.

Overall, an AI Engineer connects artificial intelligence with software engineering. The role is not simply to train models or call an LLM API. It involves designing and building complete AI-powered systems that are useful, secure, reliable, testable, and suitable for real-world use.

2. Real-world AI use cases

1. Customer Support Chatbots

AI-powered chatbots can understand customer questions and provide automated responses. They can also help route complex issues to human support agents.

2. Healthcare and Medical Analysis

AI can assist healthcare professionals by analyzing medical images, processing patient information, supporting documentation, and identifying patterns in medical data.

3. Recommendation Systems

AI analyzes user behavior and preferences to recommend products, movies, music, news, or other relevant content.

4. Fraud Detection

Machine learning models can analyze financial transactions and identify unusual patterns that may indicate fraudulent activity.

5. AI Coding Assistants

Generative AI can assist software developers with code generation, debugging, code explanation, testing, and documentation.

3. Responsibilities of an AI Engineer

1. Develop AI-Powered Applications

Design and build software applications that use AI and machine-learning capabilities to solve real-world problems.

2. Integrate AI Models and APIs

Connect AI models, LLMs, and external AI services to applications using APIs and SDKs.

3. Develop AI Services and Backend Systems

Build backend services that handle AI-related requests, data processing, model communication, and responses.

4. Work with Data

Collect, clean, transform, and prepare data required for AI applications and models.

5. Prompt Engineering
Design and improve prompts and instructions to obtain reliable and useful results from Generative AI and LLMs.
6. Model Selection and Integration
Select appropriate AI models based on the application's requirements, such as accuracy, cost, latency, and capabilities.
7. Evaluate AI Systems
Test AI outputs and measure factors such as accuracy, relevance, reliability, hallucinations, latency, and overall response quality.
8. Implement Security and Privacy Controls
Protect API keys, user information, company data, and AI systems from unauthorized access and misuse.
9. Deploy AI Applications
Deploy AI-powered applications and services to suitable development, testing, or production environments.
10. Monitor and Improve AI Applications
Monitor system performance, errors, costs, user feedback, and AI output quality, then make improvements when necessary.